Identifying Causes of Protectionism Using Double Machine Learning

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1 Proposed Topic

Why do some democracies protect domestic producers more than others? I aim to study whether factors such as exports, unemployment rates, GDP, and inflation rates causally shape how protectionist a country's trade policy is. Other analyses CITE Ehrlich and similar often test for the impact of interactions between firms and policymakers and domestic institutions on protectionism, while holding economic conditions constant, meaning that additional testing must be done in order to determine whether the economic conditions themselves also impact the degree of protectionism. Using a panel of many countries spanning several decades, I aim to estimate the effect of each economic factor using double machine learning (DML), which lets flexible learners such as random forests and neural networks estimate causal parameters in the presence of complex confounding variables (Chernozhukov et al., 2016).

This project aims to answer the question **Which domestic factors cause countries to adopt more protectionist trade policies, and how robust are these effects?** by testing a variety of hypotheses regarding protectionism.

H1 (Exports): Exports being a higher percentage of GDP decrease protectionism. Countries that export a significant amount of products abroad prefer other countries to keep their markets open, and worry that raising their own tariffs may lead to retaliation.

H2 (Unemployment): Increased unemployment rates cause increased protectionism. Free trade allows products to be imported from abroad, meaning that local industries may suffer, leading to high unemployment rates. Therefore, those suffering from unemployment and the firms forced to lay off workers will advocate for protectionism in order to maintain local jobs.

H3 (GDP per Capita): Increases in GDP per capita causes increased protectionism. need help with this one

H4 (Total GDP in US$): A higher total GDP causes increased protectionism. Small economies are unable to produce all goods, therefore imports are essential. Larger economies are able to produce more goods domestically and can afford for trade to be closed Is this right? Check if Ehrlich cited anything useful and look at that paper.

H5 (Inflation): Increased inflation rates cause decreased protectionism. Inflation means an increase in domestic prices, and cheap imports may help keep prices low maybe reword this.

For the sake of this analysis, the level of protectionism will be quantified using tariff rates. Tariffs are directly set by policymakers, meaning that changes in tariff rates directly reflect policy choices.

2 Prior Work

3 Proposed Methods and Current Progress

This is summarizing the youtube series on DML (the green one that pops up when you google double machine learning youtube). Find the FWL Theorem paper and check this info against it, then cite FWL).

Double Machine Learning (DML) is based on the Frisch-Waugh-Lovell (FWL) Theorem, which says that interactions between variables can be combatted using multiple linear regression analyses. First, regress the treatment parameter D on the independent variable you'd like to test (X_T) to find the treatment residual (r_T). Then, regress the outcome (Y) on the nuisance variables (X_N) to obtain the nuisance residual (r_N), giving you the variation of the outcome independent of X_T. Finally, regress r_N on r_T. What does this last regression give you? DML takes the FWL Theorem a step further, expanding it beyond linear relationships. DML replaces linear regression with any machine learning algorithm I believe it's usually neural networks or random forest regressors – why these two and explain them. This enables the FWL process to be used on models in which the variables are not related linearly.

Add a paragraph on how prior analyses use linear regression. Mention the assumption about linear relationships and use as justification for DML. Also I don't think Ehrlich even attempted FWL, so that's not good either.

Data on tariff rates (World Bank Group, 2026*e*), exports (World Bank Group, 2026*a*), unemployment rates (World Bank Group, 2026*f*), GDP (World Bank Group, 2026*b*,*c*), and inflation (World Bank Group, 2026*d*) and the number of political parties (Varieties of Democracy, 2026) have been obtained, and the required data reduction and cleaning is complete. A simple Python program (Harms, 2026) was written to pull the data needed from each dataset, and add the data to one large .csv file. A subset of the data is shown in table 1. figure out how to to ref command so table numbers don't get mixed up

Country	Year	% Exports	Unemp.	GDP/Capita	GDP (US$)	Inflation	No. Parties	Tariff
ALB	1997	10.04	16.874	728.55	4.71e9	33.18	2.87	14.41
ALB	2000	19.53	19.023	1160.42	6.15e9	0.05	2.87	11.65
...	...	...	...	...	...	...	...	...
ZMB	2020	46.72	6.032	951.64	2.34e10	15.73	2.78	4.82

Table 1: Dataset that will be used for double machine learning. % Exports, unemployment rate, GDP/capita, GDP in US$, inflation rate, and number of political parties are independent variables, tariff rate is the dependent variable.

Put together a model (linear model, but just to show process) and go through FWL step by step to explain. Do this just for one variable and explain the for loop. Also I hate this gap between table and next paragraph, figure out how to get rid of it.

Parameter	coef	std err	t	$P > \|t\|$	2.5 %	97.5 %
Exports	-0.0227	0.00962	-2.363	0.0181	-0.0416	-0.00388
Unemployment	0.0279	0.0511	0.546	0.585	-0.0722	0.128
GDP/Capita	-3.898	8.559e-6	-4.554	5.254e-6	-5.483e-5	-2.221e-5
GDP (US$)	-2.473e-13	0.00153	-1.610	0.107	-5.483e-13	5.375e-14
Inflation	-5.581e-4	0.00153	-0.366	0.715	-0.00355	0.00243
Number of Parties	0.0679	0.0725	0.936	0.350	-0.0743	0.210

Table 2: Preliminary results. A P-value less than 0.05 shows a statistically significant causal relationship.

4 Impacts of Protectionism

5 Notes

- Check DML Paper citation – May be linking to ArXiv, find actual publication if so
- Is JeanReshef2017 peer reviewed? Double check.
- Check years for World Bank data and V-Dem data

References

Chernozhukov, Victor, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, and Whitney Newey. 2016. "Double Machine Learning for Treatment and Causal Parameters." : Forthcoming.

Harms, Emily. 2026. "Data Cleaning Code." https://github.com/harmse1tcnj2027-arch/POL380/blob/main/DataCleaning.py.

Varieties of Democracy. 2026. "Country-Party-Date: V-Party." Varieties of Democracy.

World Bank Group. 2026*a*. "Export of goods and services (% of GDP) [NE.EXP.GNFS.ZS]." World Development Indicators.

World Bank Group. 2026*b*. "GDP (constant 2015 US$) [NY.GDP.MKTP.KD]." World Development Indicators.

World Bank Group. 2026*c*. "GDP per capita (current US$) [NY.GDP.PCAP.CD]." World Development Indicators.

World Bank Group. 2026*d*. "Inflation, consumer prices (annual %) [FP.CPI.TOTL.ZG]." World Development Indicators.

World Bank Group. 2026*e*. "Tariff rate, applied, weighted mean, all products (%) [TM.TAX.MRCH.WM.AR.ZS]." World Development Indicators.

World Bank Group. 2026*f*. "Unemployment (% of total labor force) [SL.UEM.TOTL.ZS]." World Development Indicators.